

CSCI E-106: Assignment 5

Brian Ledger

Problem 1

Refer to the CDI data set. The number of active physicians (Y) is to be regressed against total population, total personal income, land area and geographic region ("Geographic region classification is that used by the U.S. Bureau of the Census, where: 1 = NE, 2 = NC, 3 = S, 4 = W")

```
cdi_data = read.csv("CDI Data.csv")[,c(
  "Number.of.active.physicians",
  "Total.population",
  "Total.personal.income",
  "Land.area",
  "Geographic.region"
)]
tail(cdi_data)
```

```
##      Number.of.active.physicians Total.population Total.personal.income
## 435                          67          101154          1954
## 436                          98          101115          1407
## 437                         193          100900          2737
## 438                          87          100498          1323
## 439                         192          100374          1857
## 440                         122          100043          1647
##      Land.area Geographic.region
## 435         461                3
## 436         478                3
## 437         556                3
## 438         539                3
## 439        1159                4
## 440         582                3
```

a-) Create Dummy variables for Geographic Region.

```
cdi_data$Geographic.region = factor(
  cdi_data$Geographic.region,
  c(2,3,4,1),
  c(".NC", ".S", ".W", ".NE"),
)
```

b-) Fit a regression model by using all variables above.

We'll make regressions with and without geographic region.

```
summary(lm(Number.of.active.physicians ~ . -Geographic.region, cdi_data))
```

```
##
## Call:
## lm(formula = Number.of.active.physicians ~ . - Geographic.region,
```

```
##      data = cdi_data)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -1855.6   -215.2    -74.6       79.0    3689.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.332e+01  3.537e+01  -0.377  0.706719
## Total.population    8.366e-04  2.867e-04   2.918  0.003701 **
## Total.personal.income 9.413e-02  1.330e-02   7.078  5.89e-12 ***
## Land.area       -6.552e-02  1.821e-02  -3.597  0.000358 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 560.4 on 436 degrees of freedom
## Multiple R-squared:  0.9026, Adjusted R-squared:  0.902
## F-statistic: 1347 on 3 and 436 DF,  p-value: < 2.2e-16

cdi_lm = lm(Number.of.active.physicians ~ ., cdi_data)
summary(cdi_lm)

##
## Call:
## lm(formula = Number.of.active.physicians ~ ., data = cdi_data)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -1845.3   -214.2    -84.4       77.5    3693.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -3.520e+01  5.741e+01  -0.613  0.54014
## Total.population    7.984e-04  2.949e-04   2.707  0.00705 **
## Total.personal.income 9.605e-02  1.372e-02   6.999  9.89e-12 ***
## Land.area       -5.902e-02  2.112e-02  -2.794  0.00543 **
## Geographic.region.S    4.772e+01  7.085e+01   0.674  0.50095
## Geographic.region.W   -2.638e+01  9.464e+01  -0.279  0.78057
## Geographic.region.NE    1.336e+01  7.828e+01   0.171  0.86459
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 561.7 on 433 degrees of freedom
## Multiple R-squared:  0.9028, Adjusted R-squared:  0.9015
## F-statistic: 670.6 on 6 and 433 DF,  p-value: < 2.2e-16
```

Comment on your model.

The coefficients of determination are more or less equal.

None of the geographic region have a significant p-value (at $\alpha = 0.5$). We would assess that there is no relationship between geographic region and number of active physicians.

The intercept doesn't have a significant p-value. We would assess that the number of active physicians wouldn't be significantly different from zero if all variables were zero. This scenario is non-sense in the context of land area, population, etc.

c-) **Test whether any geographic effects are present; use $\alpha = .10$.**

None of the geographic areas regress with a p-value less than $\alpha = .10$.

State the alternatives, decision rule, and conclusion.

The null hypothesis is that the geographic region has no effect on number of active physicians.

The alternative hypothesis is that the geographic region has an effect on the number of active physicians.

```
alpha = 0.1
df = nrow(cdi_data)-7
t_critical = qt(1-alpha, df)
p_critical = pt(t_critical, df)
```

The critical T-value is 1.284. If the T-value of any factor is greater than the critical T-value, it would be significant.

Because all of the T-values of the geographic regions are less than the critical T-value, we do not discharge the null hypothesis. We assess that the geographic region has no effect on the number of active physicians.

What is the P-value of the test?

The critical P-value of the test is 0.9.

d-) **Check the constant variance assumption for the errors.**

The null hypothesis of the Brown Forsythe test is that the variances of all of the groups of residuals are equal.

The alternative hypothesis is that the variances of residuals differ between groups.

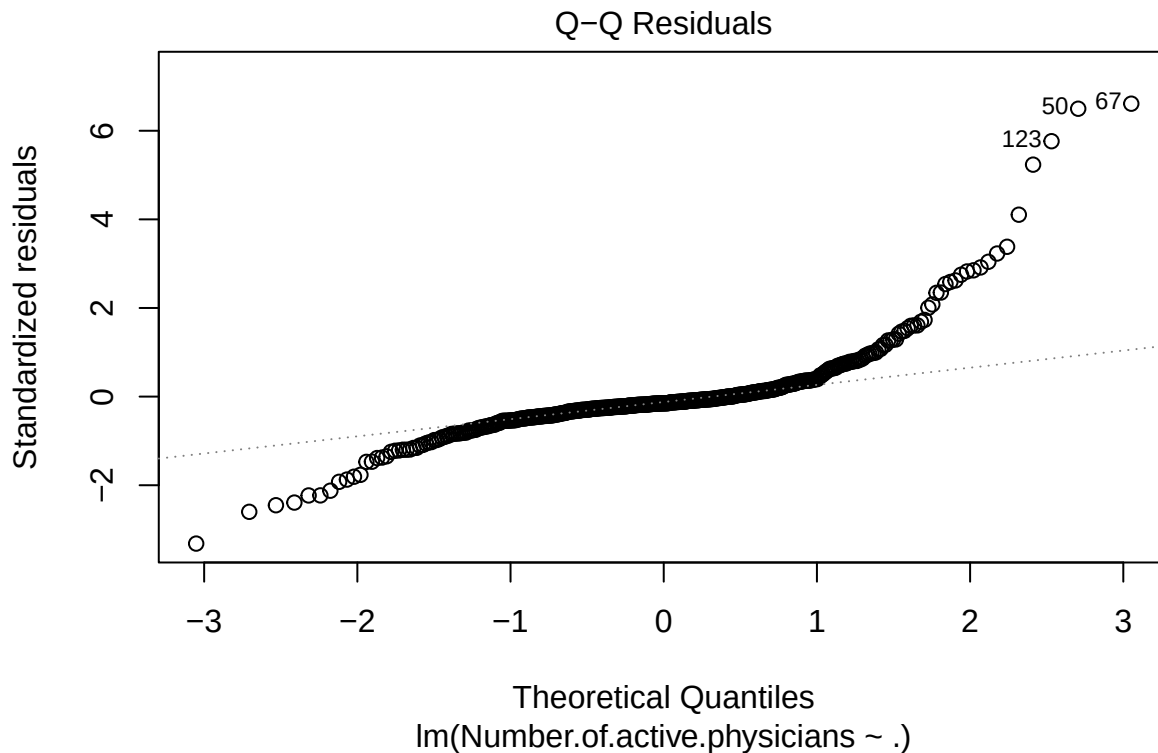
```
leveneTest(cdi_lm$residuals, cdi_lm$model$Geographic.region, center=median)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    3  0.1624 0.9217
##           436
```

The critical F-value is 0.5836886. We cannot discharge the null hypothesis. We assess that the variances of all of the groups of residuals are equal.

e-) **Check the normality assumption.**

```
plot(cdi_lm, which=2)
```



The Q-Q plot shows significant deviation at the ends of the residuals.

```
shapiro.test(cdi_lm$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  cdi_lm$residuals
## W = 0.75684, p-value < 2.2e-16
```

The Shapiro test indicates that the residuals are not normally distributed.

f-) Check for large leverage points.

The hat matrix of a linear regression is defined by:

$$\hat{H} = X(X^T X)^{-1} X^T$$

The hat matrix is a symmetric, idempotent, projection matrix that takes observed responses to the fitted values of a linear regression. The hat matrix is unique when the design matrix is non-singular.

The diagonal elements of the hat matrix are called the leverage-scores.

The sum of the leverage scores is the sum of the diagonal elements. The sum of diagonal elements of a matrix is called the trace. In the context of symmetric, idempotent matrices, the trace is equal to the rank of the matrix. Thus, when the design matrix is non-singular (full rank), the sum of the leverage scores is the same as the degrees of freedom in the design matrix.

The mean leverage score is $\frac{df}{N}$. Leverage scores can range from $\frac{1}{N}$ to 1. Scores of $\frac{1}{N}$ are constituted by points that lie on the mean, and therefore have no leverage, and no influence. Scores that approach 1 are constituted by increasingly extreme outliers, with high leverage, but not necessarily high influence.

One convention for identifying points of high leverage is to observe points in excess of 2 or 3 times the mean leverage score.

```

N = nrow(cdi_data)
cdi_df = attributes(logLik(cdi_lm))$df
cdi_mean_ls = cdi_df/N
cdi_hvs = hatvalues(cdi_lm)
cdi_hl_indices = which(cdi_hvs > 3*cdi_mean_ls)

```

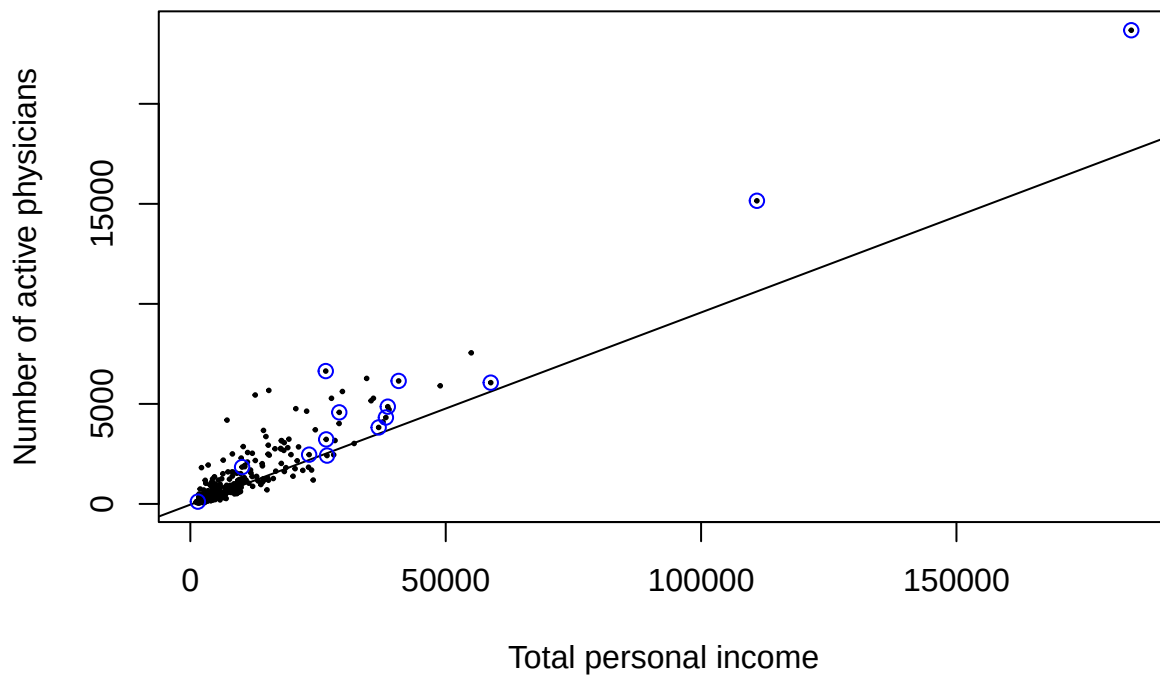
There are points in excess of 3 times the mean leverage score.

```

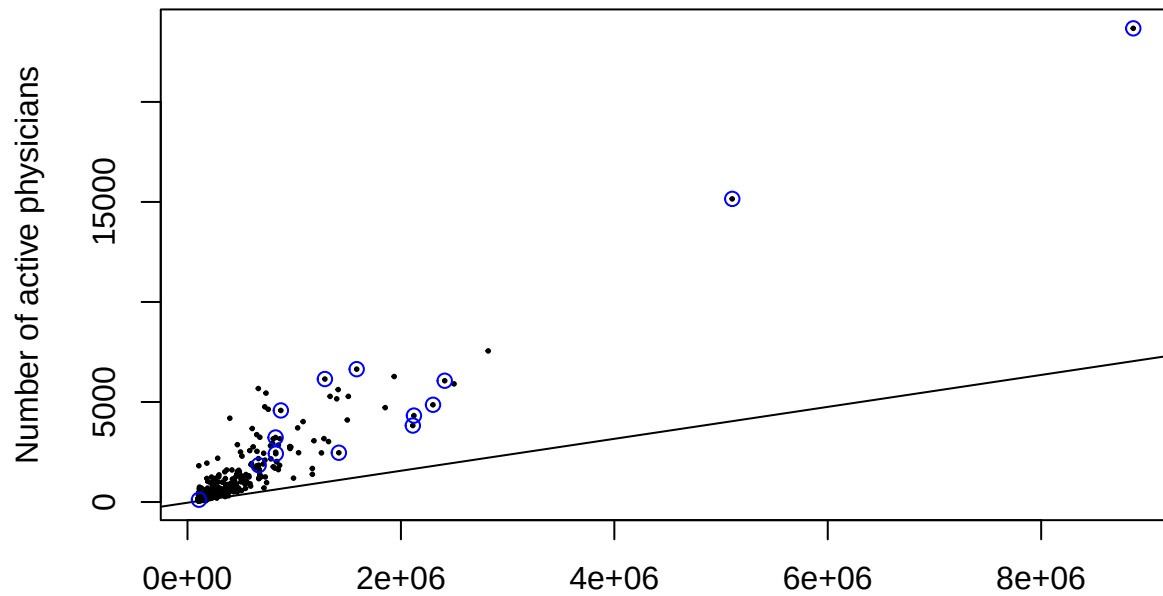
for (factor in c("Total.personal.income", "Total.population", "Land.area")) {
  factor_hum = chartr(".", " ", factor)
  plot(
    cdi_data[,factor],
    cdi_data$Number.of.active.physicians,
    main=paste("Physicians", factor_hum, sep=" vs "),
    xlab=factor_hum,
    ylab="Number of active physicians",
    pch = 19,
    cex=0.25
  )
  abline(cdi_lm$coefficients["(Intercept)"], cdi_lm$coefficients[factor])
  points(
    cdi_data[cdi_hl_indices, factor],
    cdi_data[cdi_hl_indices, "Number.of.active.physicians"],
    col="blue"
  )
}

```

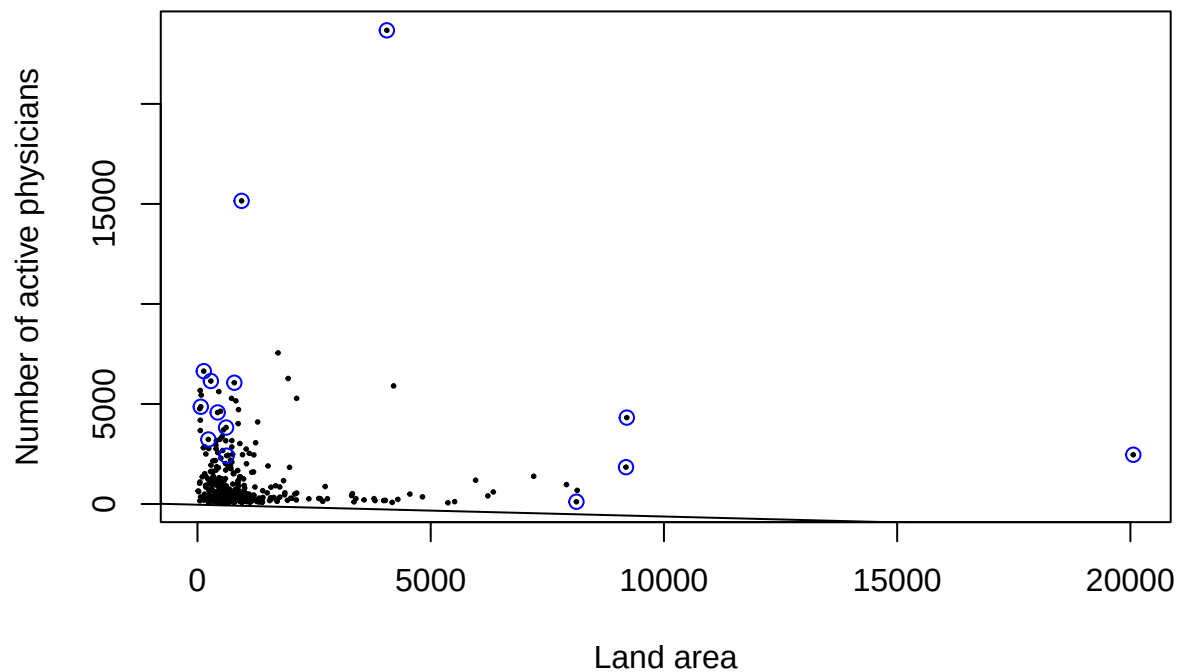
Physicians vs Total personal income



Physicians vs Total population



Physicians vs Land area



g-) Check for outliers.

```
cdi_lm_residual_mean = mean(cdi_lm$residuals)
cdi_outlier_crit = 3*sd(cdi_lm$residuals)
cdi_outlier_hi = cdi_lm_residual_mean + cdi_outlier_crit
cdi_outlier_lo = cdi_lm_residual_mean - cdi_outlier_crit
```

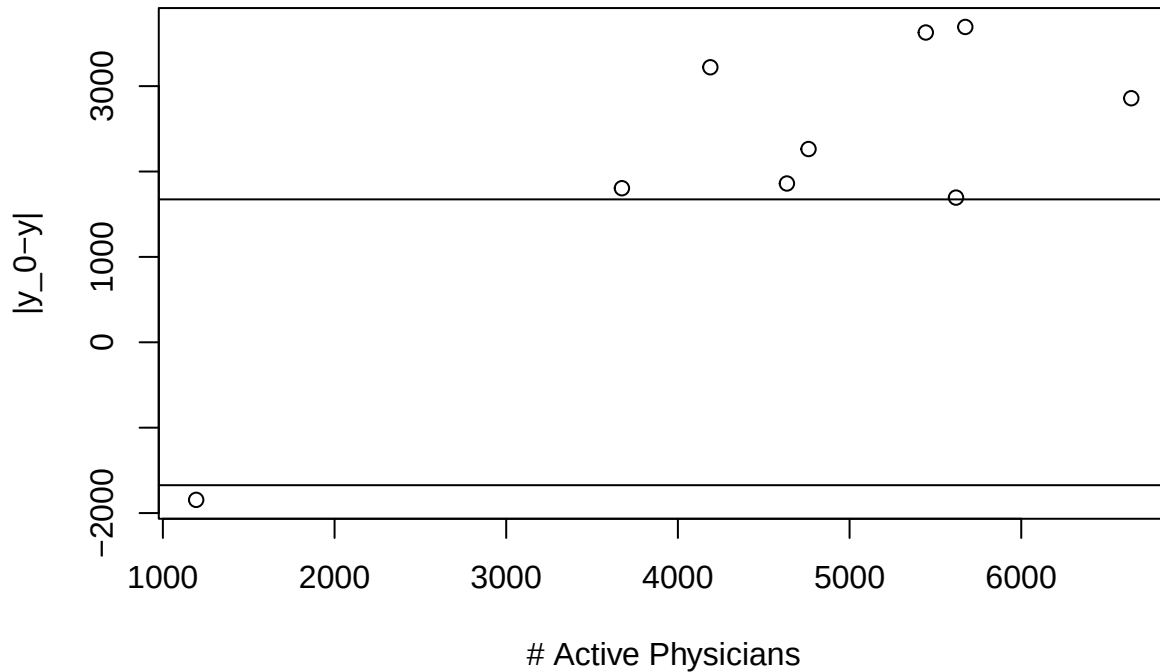
```

cdi_outlier_idx = which(cdi_lm$residuals > cdi_outlier_hi | cdi_lm$residuals < cdi_outlier_lo)

cdi_outliers = cdi_data[cdi_outlier_idx,]
cdi_outlier_residuals = cdi_lm$residuals[cdi_outlier_idx]
plot(
  cdi_outliers$Number.of.active.physicians,
  cdi_outlier_residuals,
  main = "CDI Linear Regression Residual Outliers",
  xlab = "# Active Physicians",
  ylab = "|y_0-y|",
)
abline(cdi_outlier_hi, 0)
abline(cdi_outlier_lo, 0)

```

CDI Linear Regression Residual Outliers



h-) **Check for influential points.**

DFFITS, short for difference of fits, is a measure of influence. The definition of DFFITS for a point at index i is given by

$$DFFITS_i = \frac{\hat{y}_i - \hat{y}_{(i)}}{\sqrt{MSE_{(i)}h_{ii}}}$$

In this formula, $\hat{y}_i$ is the value of the predictor at i ; h_{ii} is the i th leverage value; $\hat{y}_{(i)}$ and $MSE_{(i)}$ are, respectively, the prediction and mean-squared error at i for the subsample that omits the i th observation.

One convention for identifying points of high influence is to compare their DFFITS values to the following:

$$\sqrt{\frac{df_{(i)}}{N - df_{(i)}}}$$

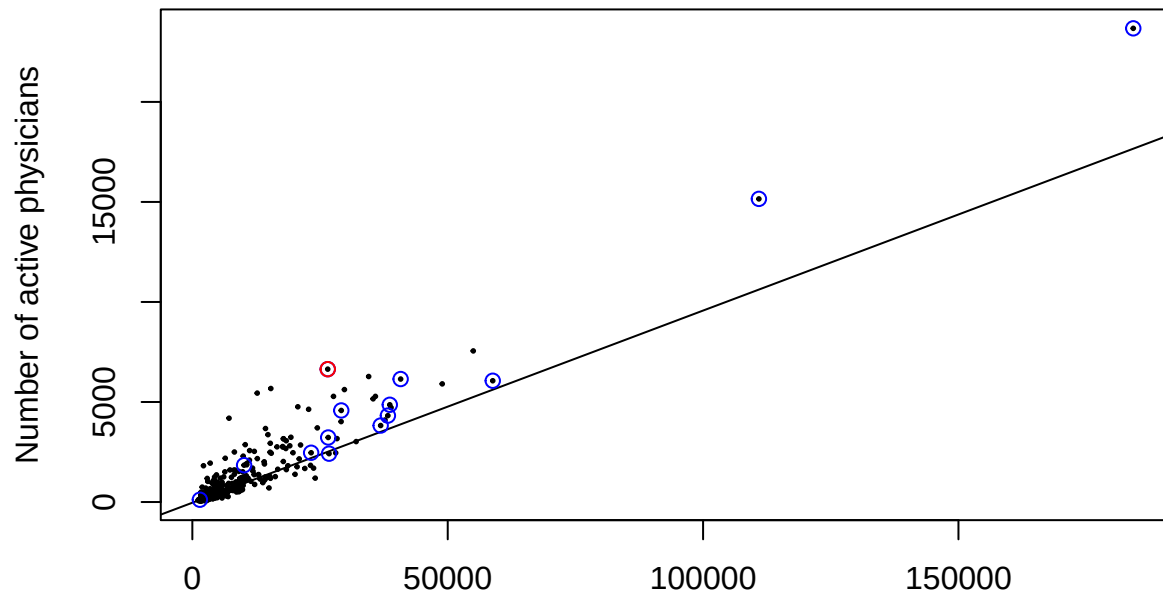
where $df_{(i)}$ is the degrees of freedom of the subsample, i.e. $df - 1$.

```
cdi_dffits = dffits(cdi_lm)
cdi_dffits_crit = sqrt((df-1)/N)
cdi_dffits_hi = which(cdi_dffits > cdi_dffits_crit)
print(cdi_data[cdi_dffits_hi,])

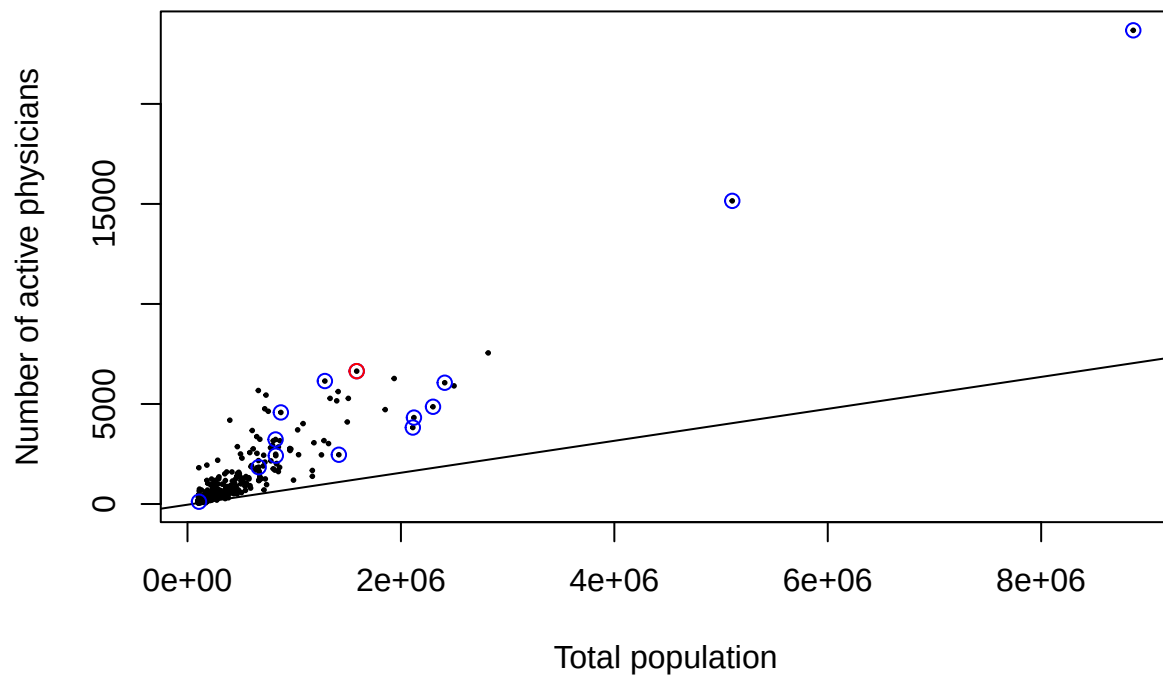
##      Number.of.active.physicians Total.population Total.personal.income Land.area
## 11                      6641          1585577          26512          135
##      Geographic.region
## 11                      .NE

for (factor in c("Total.personal.income", "Total.population", "Land.area")) {
  factor_hum = chartr(".", " ", factor)
  plot(
    cdi_data[,factor],
    cdi_data$Number.of.active.physicians,
    main=paste("Physicians", factor_hum, sep=" vs "),
    xlab=factor_hum,
    ylab="Number of active physicians",
    pch = 19,
    cex=0.25
  )
  abline(cdi_lm$coefficients["(Intercept)"], cdi_lm$coefficients[factor])
  points(
    cdi_data[cdi_hl_indices, factor],
    cdi_data[cdi_hl_indices, "Number.of.active.physicians"],
    col="blue"
  )
  points(
    cdi_data[cdi_dffits_hi, factor],
    cdi_data[cdi_dffits_hi, "Number.of.active.physicians"],
    col="red"
  )
}
```

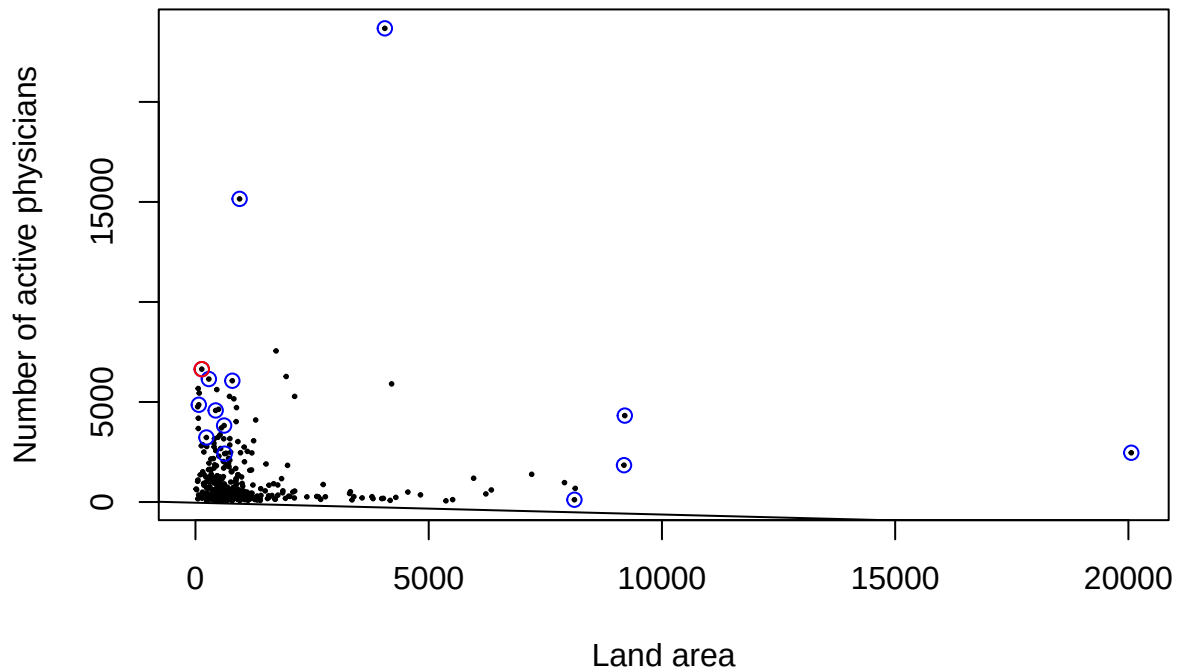
Physicians vs Total personal income



Total personal income
Physicians vs Total population



Physicians vs Land area



Problem 2

```
data(prostate, package="faraway")
p_data = prostate; rm(prostate)
tail(p_data)
```

```
##      lcavol lweight age      lbph svi      lcp gleason pgg45      lpsa
## 92 2.532903 3.6776 61 1.348073 1 -1.38629      7      15 4.12955
## 93 2.830268 3.8764 68 -1.386294 1 1.32176      7      60 4.38515
## 94 3.821004 3.8969 44 -1.386294 1 2.16905      7      40 4.68444
## 95 2.907447 3.3962 52 -1.386294 1 2.46385      7      10 5.14312
## 96 2.882564 3.7739 68 1.558145 1 1.55814      7      80 5.47751
## 97 3.471967 3.9750 68 0.438255 1 2.90417      7      20 5.58293
```

Use lpsa as the response and the other variables as predictors to answer the questions below.

a-) Create train and test data sets. Use 70set and use remaining data (30(use set.seed(1023))).

```
popN = nrow(p_data)
alphaN = round(7/10*popN)
idx_train = sample(popN, size = alphaN, replace=FALSE)
train = p_data[idx_train,]
trainN = nrow(train)
test = p_data[-idx_train,]
testN = nrow(test)
print(paste(trainN, popN, sep="/"))
```

```
## [1] "68/97"
```

Build a regression model to predict lpsa on the train data set.

```
p_lm = lm(lpsa ~ ., p_data)
```

Write down the regression model and comment on the model performance.

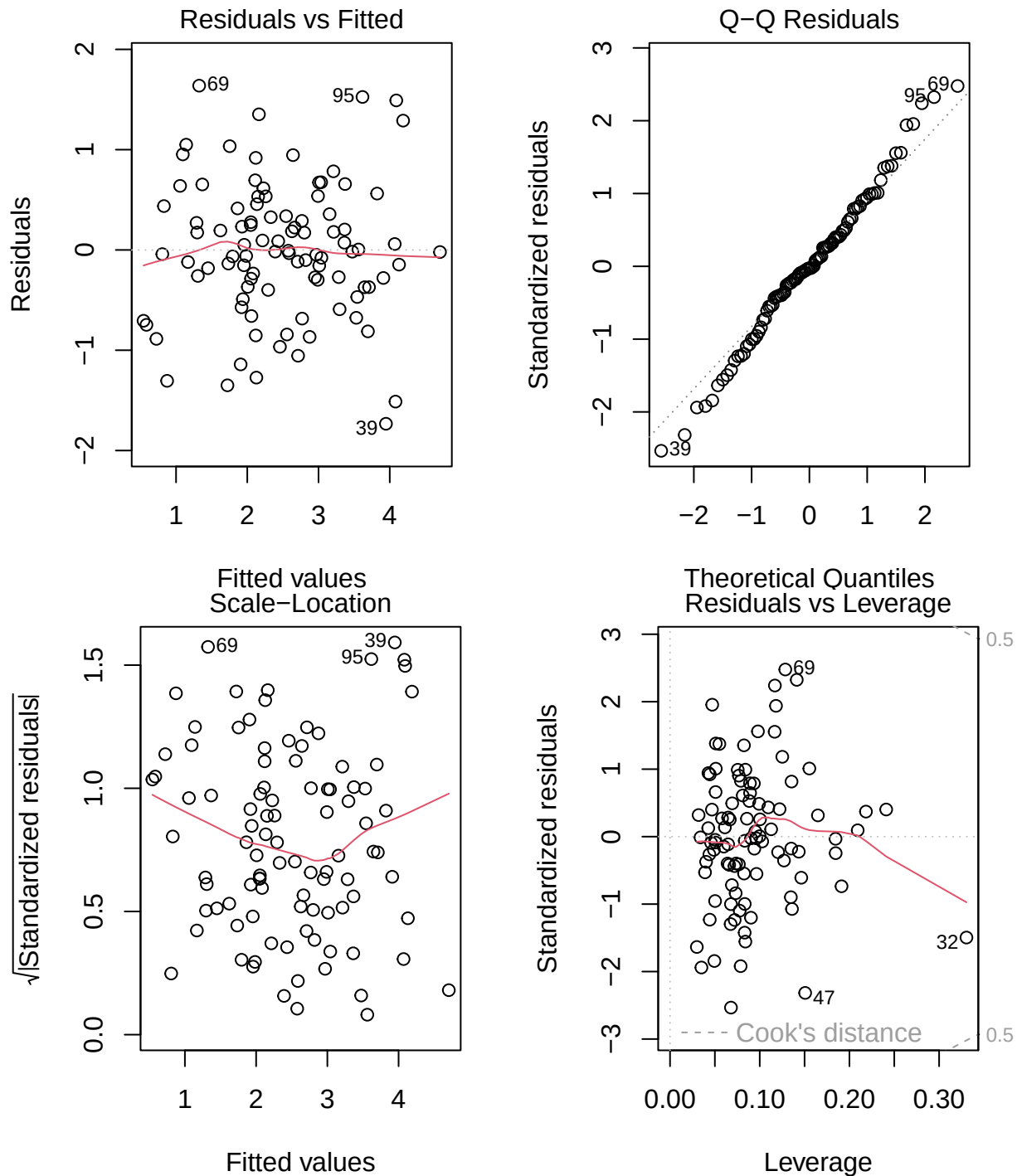
```
p_lm_summary = summary(p_lm)
print(p_lm_summary)
```

```
##
## Call:
## lm(formula = lpsa ~ ., data = p_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.7331 -0.3713 -0.0170  0.4141  1.6381
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.669337   1.296387   0.516  0.60693
## lcavol       0.587022   0.087920   6.677 2.11e-09 ***
## lweight     0.454467   0.170012   2.673  0.00896 **
## age        -0.019637   0.011173  -1.758  0.08229 .
## lbph        0.107054   0.058449   1.832  0.07040 .
## svi         0.766157   0.244309   3.136  0.00233 **
## lcp        -0.105474   0.091013  -1.159  0.24964
## gleason     0.045142   0.157465   0.287  0.77503
## pgg45       0.004525   0.004421   1.024  0.30886
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7084 on 88 degrees of freedom
## Multiple R-squared:  0.6548, Adjusted R-squared:  0.6234
## F-statistic: 20.86 on 8 and 88 DF,  p-value: < 2.2e-16
```

The factors age, lbph, lcp, gleason, and pgg45 have low significance. The Adjusted R-squared of 0.6234 might be improved by dropping these variables.

b-) Visually by using the graphs, comment on regression model assumptions.

```
par(mfrow=c(1,2))
plot(p_lm)
```



The Q-Q residuals deviate near the ends. The standardized residuals don't exceed 2 SD. There are no points in excess of 0.5 Cook's distance.

c-) Test the model performance on the test data and comment of model stability and performance.

```
predictTest <- predict(p_lm, test)
modelTest <- cbind(predictTest, test)
testResiduals = modelTest$lpsa - modelTest$predictTest
dimE = testN - p_lm$rank
```

```

sst = sum((modelTest$lpsa - mean(modelTest$lpsa))^2)
sse = sum(testResiduals^2)
ssr = sum((modelTest$predictTest - mean(modelTest$lpsa))^2)

mse = sse/dimE
rse = sqrt(mse)
rmse = sqrt(sse/length(testResiduals))
rSquared = 1.0 - (sse/sst)
mae = sum(testResiduals)/length(testResiduals)

model_summary = data.frame(MSE=c(mse), RSE=rse, RMSE=c(rmse),
                           Rsquared=c(rSquared), MAE=c(mae))
print(model_summary)

```

```

##           MSE           RSE          RMSE  Rsquared          MAE
## 1 0.3502115 0.5917867 0.4914521 0.7619369 0.08470105

```

The training and test R-squared values are 0.6548 and 0.6600 respectively. The model performs similarly on the training and test data, so it might not be overfitted.

d-) **Perform Breusch-Pagan Test, write down null and alternative hypotheses and your conclusion.**

H_0 : The error variances in the regression residuals are homoscedastic.

H_a : The error variances in the regression residuals are heteroscedastic.

```

bptest(p_lm)

```

```

##
## studentized Breusch-Pagan test
##
## data:  p_lm
## BP = 10.08, df = 8, p-value = 0.2594

```

In the linear model, the p-value is large, so we do not discharge the null hypothesis. We assess that the error variances in the regression residuals are homoscedastic.

e-) **Use the Box-Cox method to determine the best transformation on the response. if transformation is needed, perform the transformation and rebuild the model on the train data and check the model stability on the test data set.**

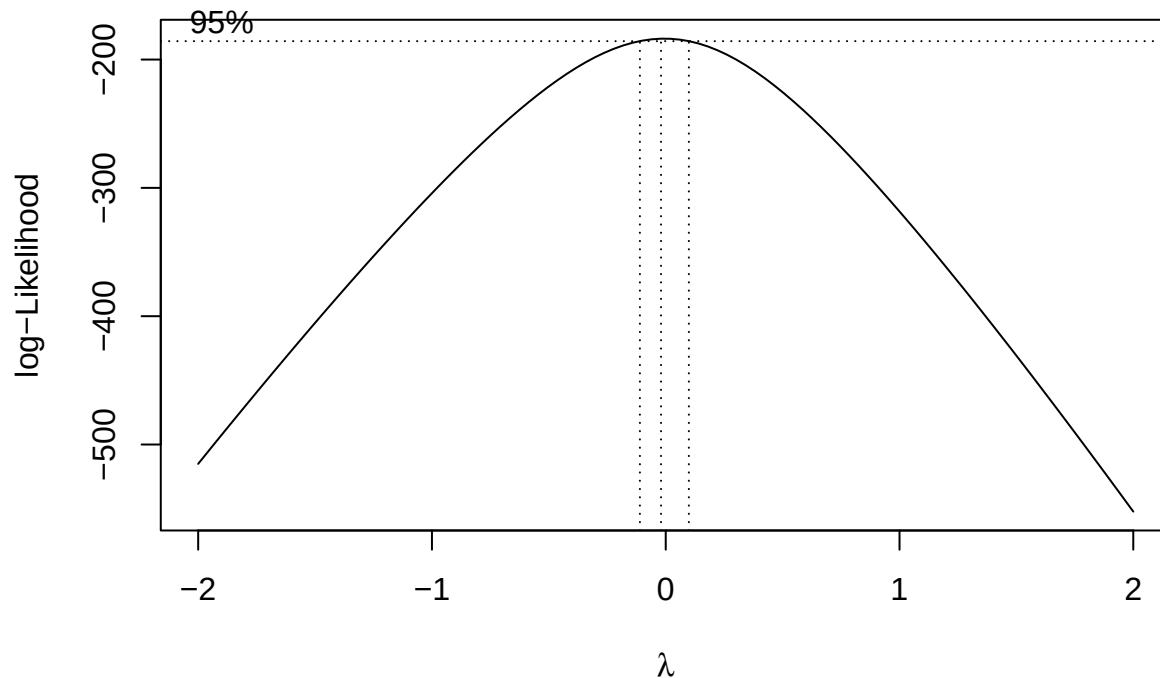
lpsa likely means log-prostate specific antigen. The log values run into the negatives, so Box-Cox can't be applied directly.

It's possible that Box-Cox was run on the raw psa data, and a value of $\lambda=0$ supported taking the log of the psa data. Let's test this.

```

p_lm_with_psa = update(p_lm, exp(lpsa) ~ .)
p_bc = MASS::boxcox(p_lm_with_psa, plotit=TRUE)

```



```
lambda_bc = p_bc$x[which.max(p_bc$y)]
```

The value of lambda is -0.020202. This supports our hypothesis.

Problem 3

Commercial properties data set.

A commercial real estate company evaluates vacancy rates, square footage, rental rates, and operating expenses for commercial properties in a large metropolitan area in order to provide clients with quantitative information upon which to make rental decisions. The data are taken from 81 suburban commercial properties include the age (X1), operating expenses and taxes (X2), vacancy rates (X3), total square footage (X4), and rental rates (Y).

a-) Fit regression model for four predictor variables to the data.

Let's look at a sample of the data.

```
cpds_data = read.csv("Commercial Properties Data Set.csv")
names(cpds_data) = c("Rate", "Age", "Costs", "Vacancy", "Size")
tail(cpds_data)
```

```
##      Rate Age Costs Vacancy   Size
## 76 15.00   3 10.34   0.00 72000
## 77 14.50   3 10.67   0.00 43404
## 78 13.50  18  8.60   0.08 59443
## 79 15.00  15 11.97   0.14 254700
## 80 15.25  11 11.27   0.03 434746
## 81 14.50  14 12.68   0.03 201930
```

```
cpds_lm = lm(Rate ~ ., cpds_data)
summary(cpds_lm)
```

```
##
## Call:
## lm(formula = Rate ~ ., data = cpds_data)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.1872 -0.5911 -0.0910  0.5579  2.9441
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.220e+01  5.780e-01  21.110 < 2e-16 ***
## Age          -1.420e-01  2.134e-02  -6.655 3.89e-09 ***
## Costs         2.820e-01  6.317e-02   4.464 2.75e-05 ***
## Vacancy       6.193e-01  1.087e+00   0.570  0.57
## Size          7.924e-06  1.385e-06   5.722 1.98e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.137 on 76 degrees of freedom
## Multiple R-squared:  0.5847, Adjusted R-squared:  0.5629
## F-statistic: 26.76 on 4 and 76 DF,  p-value: 7.272e-14
```

State the estimated regression function.

This multilinear regression has four factors: Age, Costs, Vacancy, and Size. Generally:

$$Rate = \beta_0 + \beta_1 Age + \beta_2 Costs + \beta_3 Vacancy + \beta_4 Size + \varepsilon$$

Given this model, over this data-set, we observe that β_3 is not significantly different from 0. Dropping vacancy from the model gives a marginally higher Adjusted R² (0.5667 vs 0.5629).

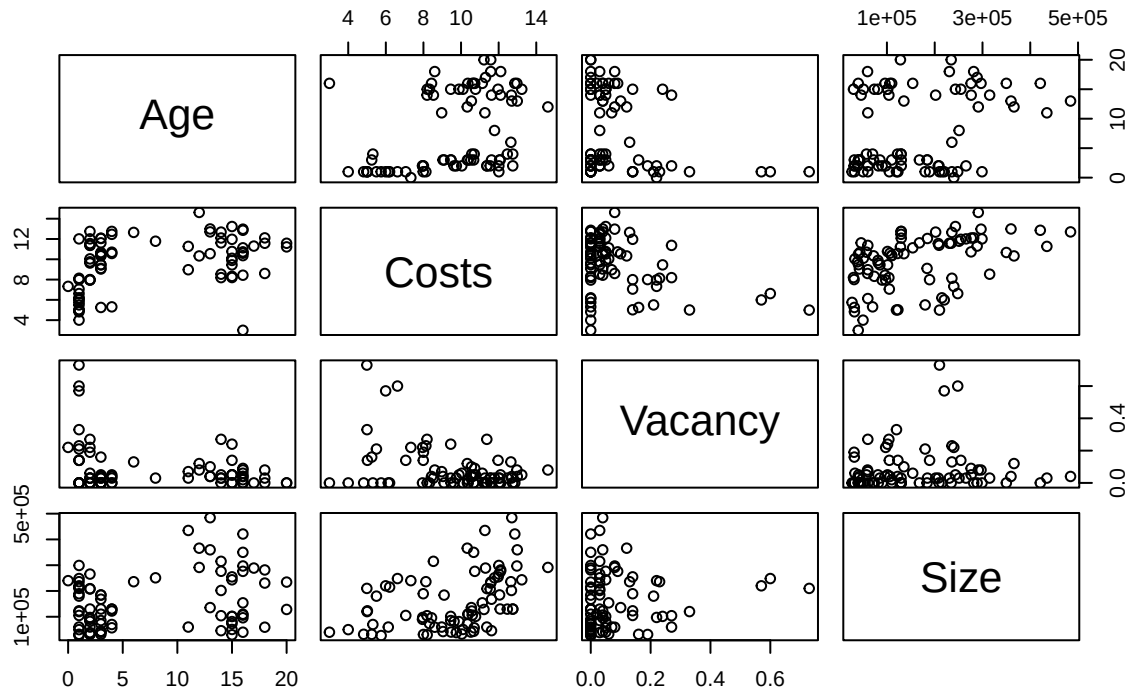
b-) Visually using the appropriate graphs, comment on the regression model assumptions.

The assumptions are:

- Observations are independent and identically distributed.
- variance is homoscedastic.

```
pairs(
  main = "check i.i.d.",
  cpds_data[,c("Age", "Costs", "Vacancy", "Size")])
```

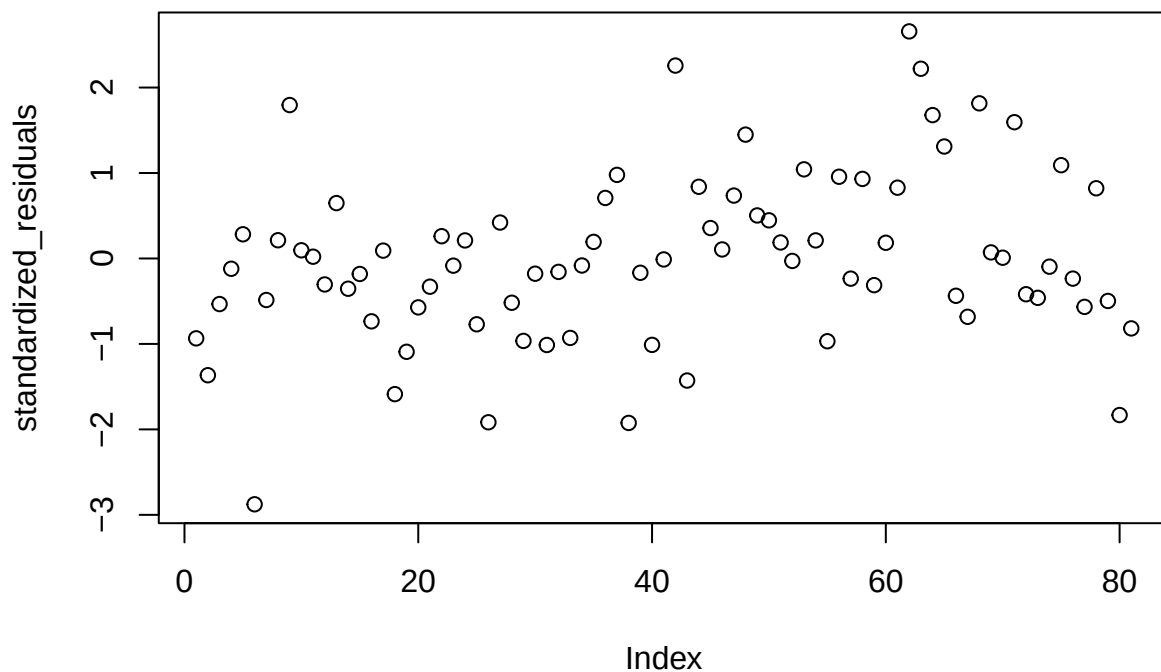
check i.i.d.



Factors Size and Cost appear to move together.

```
standardized_residuals = cpds_lm$residuals/sd(cpds_lm$residuals)
plot(standardized_residuals, main="Standardized Residuals")
```

Standardized Residuals



The residuals appear to be normally distributed. I assess that the variance is homoscedastic.

The standardized residuals hardly exceed 2 standard deviations.

c-) **Divide the 81 cases into two groups. placing the 40 cases with the smallest fitted values into group 1 and the remaining cases into group 2. Conduct the Brown-Forsythe test.**

The null hypothesis of the Brown Forsythe test is that the variances of of the two groups of fitted values are equal.

The alternative hypothesis is that the variances differ between the two groups of fitted values.

```
cpds_data_split = cpds_data
split = factor(cpds_lm$fitted.values < median(cpds_lm$fitted.values))
leveneTest(cpds_lm$fitted.values, split, center=median)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 1  1.6323 0.2051
##      79
```

The p-value for the Brown Forsythe test is 0.2051. Because it is much larger than 0.05 we cannot discharge the null hypothesis. We assess that the variances of the two groups of fitted values are equal.

d-) **Fit a second order model** ($Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4 + b_5X_1^2 + b_6X_2^2 + b_7X_3^2 + b_8X_4^2 + b_9X_1X_2 + b_{10}X_1X_3 + b_{11}X_1X_4 + b_{12}X_2X_3 + b_{13}X_2X_4 + b_{14}X_3X_4 + b_{15}X_1X_2X_3 + b_{16}X_1X_2X_4 + b_{17}X_2X_3X_4 + b_{18}X_1X_2X_3X_4 + b_{19}X_1X_3X_4$)

```
cpds_full_fit = lm(Rate ~ Age + Costs + Vacancy + Size + Age:Age + Costs:Costs + Vacancy:Vacancy + Size
```

Drop the insignificant variables from the model one at a time, refit the model and repeat this until all the variables you have in the model are significant.

```
trim_model = function(model, significance_level, runs=1) {
  model_summary = summary(model)
  high_ps = model_summary$coefficients[-1,"Pr(>|t|)"] > significance_level
  while (any(high_ps)) {
    highest_p = max(high_ps)
    highest_p_factor = names(high_ps[high_ps == highest_p])[1]
    change_formula_str = paste(".", highest_p_factor)
    model = update(model, as.formula(change_formula_str))
    model_summary = summary(model)
    high_ps = model_summary$coefficients[-1,"Pr(>|t|)"] > significance_level
  }
  model
}

summary(trim_model(cpds_full_fit, significance_level = 0.05))
```

```
##
## Call:
## lm(formula = Rate ~ Costs + Size + Size:Age + Costs:Size + Costs:Age:Vacancy +
##     Costs:Size:Age, data = cpds_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.3084 -0.4875  0.1290  0.5793  2.0553
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.110e+01  7.730e-01  14.362  < 2e-16 ***
```

```
## Costs          3.277e-01  8.267e-02   3.964 0.000168 ***
## Size           2.862e-05  5.913e-06   4.841 6.88e-06 ***
## Size:Age       -3.510e-06  5.484e-07  -6.401 1.26e-08 ***
## Costs:Size     -1.687e-06  6.139e-07  -2.748 0.007534 **
## Costs:Age:Vacancy -5.827e-02  1.754e-02  -3.323 0.001387 **
## Costs:Size:Age  2.704e-07  5.016e-08   5.392 7.99e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9759 on 74 degrees of freedom
## Multiple R-squared:  0.7021, Adjusted R-squared:  0.6779
## F-statistic: 29.06 on 6 and 74 DF,  p-value: < 2.2e-16
summary(trim_model(cpds_full_fit, significance_level = 0.01))

##
## Call:
## lm(formula = Rate ~ Size + Size:Age + Size:Age:Costs, data = cpds_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.1050 -0.5547  0.2006  0.7080  2.2417
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.387e+01  2.385e-01  58.150 < 2e-16 ***
## Size          1.367e-05  1.797e-06   7.610 5.71e-11 ***
## Size:Age      -3.392e-06  4.929e-07  -6.881 1.40e-09 ***
## Size:Age:Costs 2.441e-07  4.125e-08   5.916 8.57e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.105 on 77 degrees of freedom
## Multiple R-squared:  0.6026, Adjusted R-squared:  0.5871
## F-statistic: 38.92 on 3 and 77 DF,  p-value: 2.051e-15
```

Compare this again the model in part b)

When we trim to a 0.05 significance level, Adjusted R-Squared increases, from 0.5629 to 0.6779. When we trim to a 0.01 significance level, Adjusted R-Squared decreases to 0.4893.

The new models are significantly smaller. They are easier to comprehend.

Reducing an expanded basis in this way effectively transforms the data to a form where non-trivial relationships between the factors are accounted for, and multi-collinearity is reduced.